

## Lab 2: Create a Git Repository and Develop Features Using Multiple Branches

### Initial Project Structure

Assume the previous folder is FEDFSEC60, which already contains a Git repository and a project named firstproject.

FEDFSEC60/

```
|
|
|— .git/
|— firstproject/
|  └─ index.html
```

### Sample index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>First Project</title>
</head>
<body>
  <h1>Welcome to My First Project</h1>
  <p>This project demonstrates Git branching and merging.</p>
</body>
</html>
```

### Part 1: Open the Existing Project

#### Step 1: Open previous project folder

Open Command Prompt (CMD) inside the FEDFSEC60 folder.

Example:

```
C:\Users\Student\Documents\FEDFSEC60>
```

#### Step 2: Navigate to the project folder

```
cd firstproject
```

### Part 2: Create README and First Commit

#### Step 3: Create README.md

```
echo # My Project > README.md
```

Project structure:

```
firstproject/
|— index.html
└─ README.md
```

#### Step 4: Check Git status

```
git status
```

Expected:

Untracked files:

README.md

### Step 5: Add README to staging

```
git add .
```

### Step 6: Commit changes

```
git commit -m "readme file created"
```

Expected:

[main abc123] readme file created

1 file changed

## Part 3: Check Remote Repository

### Step 7: Verify remote origin

```
git remote -v
```

#### If output exists

origin https://github.com/username/firstproject.git (fetch)

origin https://github.com/username/firstproject.git (push)

Then push:

```
git push origin main
```

#### If no remote exists

Add remote:

```
git remote add origin https://github.com/bhaskarmarapelli/firstproject.git  
git push -u origin main
```

## Part 4: Create a Feature Branch

### Step 8: Check current branch

```
git branch
```

Output:

\* main

### Step 9: Create branch feature1

```
git branch feature1
```

### Step 10: Switch to feature1

```
git switch feature1
```

Output:

Switched to branch 'feature1'

## Part 5: Develop a Login Page

### Step 11: Create login.html

```
notepad login.html
```

Paste:

```
HTML
<!DOCTYPE html>
<html>
<head>
  <title>Login</title>
</head>
<body>
  <h2>Login Page</h2>

  <form>
    Username:
    <input type="text"><br><br>

    Password:
    <input type="password"><br><br>

    <button>Login</button>
  </form>
</body>
</html>
```

Save and close.

Project structure:

```
firstproject/
├── index.html
├── login.html
└── README.md
```

### Step 12: Stage the file

```
git add .
```

### Step 13: Commit

```
git commit -m "login page added"
```

## Part 6: Merge Feature1 into Main

### Step 14: Switch to main

```
git switch main
```

### Step 15: Merge feature1

```
git merge feature1
```

Output:

Updating ...

Fast-forward

login.html added

Project structure after merge:

firstproject/

├── index.html

├── login.html

└── README.md

### Step 16: Push to GitHub

```
git push origin main
```

## Part 7: Create Another Feature Branch

### Step 17: Create feature2

```
git branch feature2
```

### Step 18: Switch to feature2

```
git switch feature2
```

### Step 19: Edit index.html

```
notepad index.html
```

Add a login link:

```
<!DOCTYPE html>
<html>
<head>
  <title>First Project</title>
</head>
<body>
  <h1>Welcome to My First Project</h1>
  <p>This project demonstrates Git branching and merging.</p>

  <a href="login.html">Login Page</a>
</body>
</html>
```

Save.

### Step 20: Stage changes

```
git add .
```

### Step 21: Commit

```
git commit -m "login link added"
```

### Part 8: Merge Feature2 into Main

#### Step 22: Switch to main

```
git switch main
```

#### Step 23: Merge feature2

```
git merge feature2
```

#### Step 24: Push

```
git push origin main
```

At this point the website contains:

- Home page
- Login page
- Link from home page to login page

### Part 9: Create a Merge Conflict

This is the most important classroom demonstration.

#### Step 25: Switch to feature1

```
git switch feature1
```

#### Step 26: Edit README

```
notepad README.md
```

Change to:

```
# My Project
```

```
This line was added in feature1.
```

Save.

#### Step 27: Stage

```
git add .
```

#### Step 28: Commit

```
git commit -m "edited readme in feature1"
```

#### Step 29: Switch to main

```
git switch main
```

### Step 30: Edit README differently

```
notepad README.md
```

Change to:

```
# My Project
```

```
This line was added in main branch.
```

Save.

### Step 31: Stage

```
git add .
```

### Step 32: Commit

```
git commit -m "edited readme in main"
```

## Part 10: Merge and Resolve Conflict

### Step 33: Merge feature1

```
git merge feature1
```

Git output:

Auto-merging README.md

CONFLICT (content): Merge conflict in README.md

Automatic merge failed; fix conflicts and then commit the result.

### Step 34: Open README

```
notepad README.md
```

Git inserts conflict markers:

```
# My Project
```

```
<<<<<<< HEAD
```

```
This line was added in main branch.
```

```
=====
```

```
This line was added in feature1.
```

```
>>>>>> feature1
```

### Resolve the conflict

Keep both lines:

```
# My Project
```

```
This line was added in main branch.
```

```
This line was added in feature1.
```

Save.

### Step 35: Stage the resolved file

```
git add .
```

### Step 36: Commit the resolution

```
git commit -m "conflict resolved"
```

Git creates a merge commit.

### Step 37: Verify history

```
git log --oneline --graph --all
```

Example:

```
* e3f4g5 (HEAD -> main) conflict resolved
|
| * a1b2c3 (feature1) edited readme in feature1
|
| * | d4e5f6 edited readme in main
|/
* 789abc login link added
* 456def login page added
* 123abc readme file created
```

### Final Project Structure

FEDFSEC60/

```
└─ firstproject/
   └─ index.html
   └─ login.html
   └─ README.md
└─ .git/
```

### Final README.md

# My Project

This line was added in main branch.

This line was added in feature1.

### Final index.html

```
<!DOCTYPE html>

<html>
```

```
<head>
  <title>First Project</title>
</head>
<body>
  <h1>Welcome to My First Project</h1>
  <p>This project demonstrates Git branching and merging.</p>

  <a href="login.html">Login Page</a>
</body>
</html>
```